

LETTER

Urbanisation tolerance and the loss of avian diversity

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Abstract

Urbanisation is considered an important driver of current biodiversity loss, but the underlying causes are not fully understood. It is generally assumed that this loss reflects the fact that most organisms do not tolerate well the environmental alterations associated with urbanisation. Nevertheless, current evidence is inconclusive and the alternative that the biodiversity loss is the result of random mechanisms has never been evaluated. Analysing changes in abundance between urbanised environments and their non-urbanised surroundings of > 800 avian species from five continents, we show here that although random processes account for part of the species loss associated with urbanisation, much of the loss is associated with a lack of appropriate adaptations of most species for exploiting resources and avoiding risks of the urban environments. These findings have important conservation implications because the extinction of species with particular features should have higher impact on biodiversity and ecosystem function than a random loss.

Keywords

Brood value, extinction risk, life history, neutral theory, rapid human-induced environmental changes.

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INTRODUCTION

Urbanisation is one of the most pervasive and extreme forms of environmental alteration, posing a major threat to biodiversity and altering fundamental ecosystem services upon which human civilisation depends (McKinney 2002, 2006; McDonnell & Hahs 2008; Shochat *et al.* 2010; Aronson *et al.* 2014). Although the impact of urbanisation varies with the speed and intensity of environmental alterations, a common consequence is a loss of biodiversity (McKinney 2002; Clergeau *et al.* 2006; Aronson *et al.* 2014; but see Desrochers *et al.* 2011). Surprisingly, however, the causes underlying this loss remain little understood (Shochat *et al.* 2006, 2010). Given that urban land cover is predicted to increase by 1.2 million km² in the first 30 years of the 21st century (Seto *et al.* 2012), understanding how the urbanisation process will impact biodiversity is currently a major priority for both conservation biology and human wellbeing.

The loss of biodiversity associated with urbanisation is often assumed to reflect the fact that only a few species have the adaptations that confer tolerance to such rapid human-induced environmental alterations (Sih *et al.* 2011). Empirical support for this ‘urbanisation tolerance hypothesis’, as we refer to it here, comes from a growing number of studies reporting that species that occur in cities are characterised by certain biological traits assumed to allow resource use and/or hazard avoidance in such environments (Bonier *et al.* 2007; Kark *et al.* 2007; Croci *et al.* 2008; Møller 2009; Evans *et al.* 2011; Maklakov *et al.* 2011). However, most of these studies are either localised in extent, allowing little generalisation, and/or suffer from important methodological shortcomings

(see Evans *et al.* 2011; Sol *et al.* 2013a). A particularly important shortcoming is measuring tolerance to urbanisation by categorising species according to whether they occur in urban habitats or not. This assumes that presence equates to tolerance, which is unlikely to be true, and ignores that a species can be absent from a city simply because it is too scarce in the surrounding habitats to be able to establish itself there (Evans *et al.* 2011; Sol *et al.* 2013a). In addition, although many characteristics commonly believed to affect tolerance to urbanisation are typically shared by phylogenetically related species (Bennett & Owens 2002; Sol 2007; Sol *et al.* 2012), no study to date has been able to detect phylogenetic signal in the manner in which species respond to urbanisation (Evans *et al.* 2011).

Confidence in the ‘urbanisation tolerance hypothesis’ is not only undermined by a lack of firm empirical support, but a loss of species is also expected under the hypothesis of ‘functional equivalence’, where species are considered identical in terms of their birth, death and/or dispersal rates (Hubbell 2005; Ostling 2005). Two non-mutually exclusive mechanisms can be involved here. The first is a decrease in community size (i.e. the number of individuals of all species that a given locality can maintain) in urban habitats relative to less disturbed habitats, which should increase the risk of extinction of the rarest species by demographic stochasticity and Allee effects (Kaspari *et al.* 2003; Evans *et al.* 2005; Carnicer *et al.* 2007; Chiari *et al.* 2010). Although the decrease in community size depends on the area being urbanised, the importance of this ‘abundance–extinction’ mechanism is contradicted by growing evidence that urban centres often exhibit higher (not lower) population densities when compared with wildlands

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(e.g. Shochat *et al.* 2010; Møller *et al.* 2012; but see Chiari *et al.* 2010). More credence is nonetheless available for the second mechanism, a random dispersal-assembly process (Hubbell 2005; Ostling 2005). If urban communities are assembled mainly by the random dispersal of individuals from the surrounding non-urbanised environments (Ostling 2005), then some species will be absent simply because they are too scarce in the surrounding habitats to colonise and persist in urbanised habitats. Although there is evidence that urban communities are influenced by the diversity in adjacent landscapes (Clergeau *et al.* 2001; Møller *et al.* 2012), the 'functional equivalence' hypothesis has never been considered in the context of current biodiversity loss.

Because the extinction of species with particular features is expected to have a higher impact on biodiversity than a random loss (Isaac *et al.* 2007), our insufficient understanding of the 'urbanisation tolerance' and 'functional-equivalence' hypotheses has precluded a deeper assessment of the conservation implications of the urbanisation process. Although contrasting both hypotheses is difficult due to the idiosyncrasies typical of community studies, important insight can be gained by combining geographically replicated local studies with null community models derived from explicit definitions of species pools (Lessard *et al.* 2012). Here, we use such an integrative approach to assess the relative importance of each hypothesis in explaining the decline in biodiversity generally observed in urbanised environments.

We compiled published and unpublished data on bird abundances in 22 regions of five continents for which intensive surveys were available in well-defined urbanised environments, as well as in the surrounding non-urbanised habitats representing local species pools. Taking advantage of the recent publication of the complete phylogeny of birds (Jetz *et al.* 2012) and new phylogenetic Bayesian methods that allow the integration of global scale information originated at fine scale resolution (Hadfield & Nakagawa 2010), we show that although the key premises of the random dispersal-assembly process are valid, the loss of diversity explained by such a random process is, according to community simulations, modest compared to that actually observed. Support for the general importance of the 'urbanisation tolerance hypothesis' is nonetheless stronger. We show this by estimating tolerance of species to urbanisation through community simulations. Assuming that organisms decrease their densities in the less favourable habitats (Evans *et al.* 2011), we classified species as either urban avoiders or exploiters (*sensu* Blair 1996) according to whether they are, respectively, less or more abundant than expected by chance. We then used this measure to test the predictions that (1) species show consistent tolerance to urbanisation (i.e. they are consistently classified as avoider or exploiter across different regions), (2) closely related species exhibit similar levels of tolerance to urbanisation and (3) the tolerance to urbanisation can be predicted by certain features of species. We address this last prediction by testing all major hypotheses proposed to explain the varying tolerance of birds to urbanisation (Evans *et al.* 2011; Sol *et al.* 2013a). Having found evidence for all the above predictions, we finally ask to what extent the diversity loss associated with urbanisation can be explained by a low tolerance to urbanisation.

MATERIALS AND METHODS

Study system

Birds are particularly suited for the research proposed because they are frequent colonisers of urban areas (Chiari *et al.* 2010) and their life history, ecology and phylogeny are well documented. Moreover, although birds are major providers of ecosystem services, contributing to the four types of services recognised by the UN Millennium Ecosystem Assessment (Whelan *et al.* 2008), the global number of individual birds has experienced a major reduction since 1500 AD (Gaston *et al.* 2003).

Survey data

We compiled a large data set of bird abundances per unit area or unit time spent observing in 22 well-surveyed regions (locations, hereafter) containing habitats in varying degrees of urbanisation and spanning a broad geographic range (Fig. 1, see Appendices S1 and S2). Information came from published studies and our own surveys in Newcastle (Australia) and Valdivia (Chile) (see Appendix S2). In six locations, surveys were replicated across years or seasons, allowing us to estimate repeatabilities. Although some studies also reported information for exotic species, these species were not considered in this study. The analyses included 1548 estimates of abundance belonging to 842 species.

Habitat descriptions

To define the degree of urbanisation, we differentiated (1) urban habitats ('highly urbanised' environments, hereafter), where buildings are densely packed and parks are small or absent, and (2) suburbs ('moderately urbanised' environments), which are residential areas with single-family houses, commonly with backyards (Marzluff *et al.* 2001). The surrounding habitats considered as potential sources of immigrants included rural and wildland environments (See Appendices S1 and S2). While the survey method varied across studies, the same survey method was used for each habitat within each location, making data comparable. Although the comparison among habitats raises issues about detectability, the results for studies that control for detectability are consistent with the general conclusions (See Tables S1–S4).

Estimating species diversity loss with urbanisation

Because differences in richness between habitats may simply reflect differences in the number of sampled individuals, we compared species richness separately for each location with individual-based rarefactions. Changes in rarefied diversity among habitats were evaluated with a Gaussian linear mixed model, using a Bayesian approximation in the R-package 'MCMCglmm' (Hadfield 2010). The survey method and whether detectability was or not taken into account were included as fixed effects, along with the altitude and distance to Equator of the location. Location was included as a random effect, which allowed the intercept of the lines relating the response variable and habitat to vary across locations.

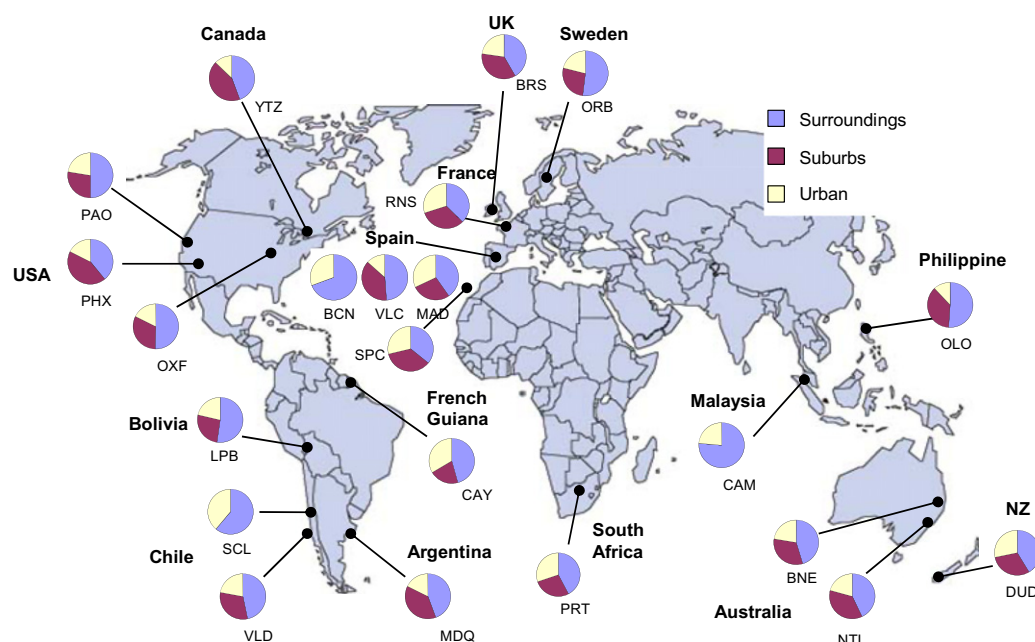


Figure 1 Worldwide location of the study areas, showing the changes in avian species richness with increasing urbanisation. The values have been rarefied to take into account differences in the number of individuals observed in each habitat. The abbreviations are as follows: BCN = Barcelona, BNE = Brisbane, BRS = Bristol, CHG = Cameron Highlands, CAY = Cayenne, DUD = Dunedin, SPC = La Palma, LPB = La Paz, MAD = Madrid, MDQ = Mar del Plata, NTL = Newcastle, OLO = Olongapo, ORB = Orebro, OXF = Oxford, PAO = Palo Alto, PHX = Phoenix, PRT = Pretoria, RNS = Rennes, SCL = Santiago de Chile, YTZ = Toronto, ZAL = Valdivia, and VLC = Valencia. Note that not all the locations contain all the habitats.

Sampling effort was also included as random factor, as suggested by Hadfield (2010). The structure and priors of all models are presented in the Appendix S3.

Testing the random dispersal-assembly hypothesis

This hypothesis was tested by assessing whether the abundance of species in urbanised environments could be predicted by their relative abundance in the surrounding rural and/or natural habitats (Clergeau *et al.* 2001). The absolute abundance of each species in the urbanised habitat (response variable) was compared with its relative abundance (i.e. the abundance of the species, divided by the total abundance of all species combined) in the surrounding environments with the MCMCglmm approach (Hadfield 2010), including location, species identity and phylogeny as random factors. In modelling the abundance of individuals in urban habitats, we detected far more zeros than would be expected for a Poisson distribution. We tackled this difficulty by using Zero Inflated Poisson models (ZIP) where the probability of measuring a zero is modelled with a binomial process and the count process (including the true zeros) is modelled by a Poisson process.

Assessing species loss due to random processes

To assess species loss due to random dispersal, for each location we created 999 random communities by randomly drawing (with replacement) individuals from a pool of species that occurred in the same proportion as in the surrounding community. To facilitate comparison and avoid confounding random dispersal with community size, the simulated commu-

nities had the same community size as the corresponding urban community. A species was considered to be absent in the urban habitat as a result of random dispersal if its abundance in all the simulations had a mode of zero.

Discerning between urban exploiters and urban avoiders

We also used the community simulations to assess whether some species were less or more abundant in urbanised environments than expected by random dispersal from the surrounding environments. Based on the null distribution of abundances in all the random communities, we considered a species as an 'avoider' if the observed abundance in the urban habitat was equal to or lower than the 5th percentile of the random abundances, an 'exploiter' if this abundance was equal to or higher than the 95th percentile, and a 'neutral' if it was observed as expected by chance. For formal analyses, we excluded the 'neutrals' because this category includes species for which information is insufficient to discern whether their populations are or are not affected by urbanisation (Sol *et al.* 2013a). Our measure of tolerance to urbanisation assumes that all species within each category have an equivalent response to urbanisation, which is unlikely to be true (Evans *et al.* 2011). Although this is inevitable, given the lack of information on population dynamics for most species, we tried to mitigate this by restricting the analyses to highly urbanised environments; hence, our metric measures whether or not a species is able to proliferate under conditions where natural habitats have heavily been modified by humans. This excludes as exploiters those species that persist in urbanised environments by exploiting more 'natural habitats' like backyards, parks and gardens.

Testing the tolerance hypothesis by quantifying repeatabilities and phylogenetic signal

We estimated species repeatability and phylogenetic signal by using the MCMCglmm approach outlined before. The binary response variable (i.e. urban avoider or urban exploiter in a particular location) was modelled with a categorical error structure, and location and species identity were included as random factors. We measured repeatability with the intraclass correlation coefficient, which in this context is the expected correlation for observations of the same species. For this analysis we only used species occurring in more than one location from different countries ($n = 72$ species). To quantify phylogenetic signal, we used the entire data set (i.e. added species that only occurred in a single location, $n = 432$ species) and included the phylogenetic relatedness among species as random effect.

Measuring traits associated with avoiders and exploiters

Based on a literature review, we collected species-level information from scientific journals, field guides and previously compiled data sets (See Appendix S4) for a number of traits that theory suggests that could influence establishment both in novel and/or urbanised environments (Evans *et al.* 2011; Sol *et al.* 2013a). These included, first, a set of life-history traits that were used to estimate the fast–slow continuum (estimated with a factor analysis including fecundity, age at first breeding, egg size and length of the incubation period), the maximum rate of population growth (R_{max} , estimated with the variables fecundity, age at first breeding and lifespan by solving Cole's equation) and the brood value (expressed as $\log_{10} \{1/[(\text{broods per year}) \times \text{lifespan}]\}$) and describing the extent to which a species concentrates the reproductive effort into a few or many events (Sol *et al.* 2012). Second, we measured variables related to the use of habitats and resources. We obtained information on the diet and habitat of species expressed as the frequency of use (not used = 0, occasionally used = 0.1, frequently used = 0.5 and almost exclusively used = 1) of a number of categories (habitat: forest, woodland, open vegetation, including tundra and grassland, and aquatic, including marshes and wetlands; resources: grain and seeds, grasses and herbs, fruits and berries, pollen and nectar, vegetative material, invertebrate, vertebrate and carrion). With this information, we calculated the niche breadth for both diet and habitat with Rao's quadratic entropy as implemented in *Resniche* (De Cáceres *et al.* 2011). In addition, we also used geographic range size in the analyses, which has been related to broader environmental tolerances (Bonier *et al.* 2007), measured as the latitudinal degrees between the North and South limits of the species' breeding distribution. Finally, we recorded a number of intrinsic attributes of species that define the way species interact with their environment. These included: (1) migratory status (i.e. resident vs. migratory in the study region); (2) relative brain size (residual of a log-log regression of brain size against body mass), as a measure of behavioural flexibility; (3) social behaviour (colonial, semicolonial/facultative and solitary breeding behaviour); (4) nesting site (ground, holes and others); (5) body mass, in grams; and (6) mating system (polygamous vs. monogamous).

Testing the tolerance hypothesis by identifying traits associated with avoiders and exploiters

We first used a model selection procedure including all the combinations of variables by using the *R* package *MuMin* (Barton 2013). For this analysis, we used *lme4* package (Bates & Martin 2009), because the information criteria used in Bayesian approaches (DIC) are considered less reliable than the Akaike information criterion (AICc) for non-Gaussian response variables. To model urban tolerance (i.e. avoiders vs. exploiters) as a function of the traits described above, we defined models with a binomial errors structure and location and taxonomy (species nested within families) as random effects. The variables that appeared in the best models (i.e. models with $\Delta\text{AICc} < 2$ relative to the model with lower AIC) were then used in formal analyses with a phylogenetic MCMCglmm (see also Appendix S3). Thus, the binary response variable was modelled with a categorical error structure, and location, species and phylogeny were included as random factors. In the best model, only those predictors that were significantly associated with the response variable were retained. We repeated the analyses with a more restrictive definition of urban avoider in which we only considered the subset of avoiders that were never observed in the urban habitat during the intensive surveys ('strict urban avoiders', hereafter).

Estimating species loss due to varying tolerance to human-altered habitats

We evaluated this by extracting the number of strict urban avoiders from the total of species present in the surrounding natural environments.

Phylogenetic information

The phylogenies in all the analyses were those recently proposed by Jetz *et al.* (2012). To take into account uncertainties in phylogenetic reconstructions, we repeated all phylogenetic-based analyses with 50 phylogenies randomly sampled from the posterior distribution of phylogenetic trees. The phylogeny had negligible effects on the conclusions, and hence we present the results with one of the randomly selected phylogenies.

RESULTS

Species richness declined with urbanisation relative to the surrounding environments in most of the locations studied (Figs. 1, S1; Table S1). The decline was more pronounced when examining highly urbanised environments (i.e. city centres), where the median decline in species richness was 52.2% (range 19.5–76.5%; *MCMCglmm*, $p\text{MCMC} < 0.001$). The decline was also clear, although less pronounced, for moderately urbanised environments (median = 27.4%, range –10.9 to 55%; *MCMCglmm*, $p\text{MCMC} < 0.001$; Table S1). These patterns do not simply reflect a greater variety of habitats in the surrounding environments because species richness also declined when the analysis was conducted at the habitat level (Table S2).

As predicted by the functional equivalence hypothesis, species abundance in both moderately (*MCMCglmm*: posterior median = 0.668, credible interval (CI) = 0.596 to 0.735, $pMCMC < 0.001$) and highly urbanised (*MCMCglmm*: posterior median = 0.788, CI = 0.691 to 0.898, $pMCMC < 0.001$) environments positively covaried with the relative abundance of species in the surroundings (Figs 2, S2 and S3; Tables S3 and S4). However, the simulations indicated that the loss of species by random dispersal only was modest compared with that actually observed (Fig. 3a and b).

In addition, the simulations revealed the existence of some species that were significantly less abundant in highly urbanised environments than expected by chance and others that were more abundant (Fig. 3c and d). While this alone does not necessarily imply that species vary in tolerance to urbanisation, the analyses of species occurring in more than one location revealed that species were consistent in the way they respond to urbanisation across regions, with some being consistently classified as avoiders and others as exploiters (repeatability = 0.652, CI = 0.506 to 0.826). We also found strong evidence for phylogenetic signal (phylogenetic heritability sensu Hadfield & Nakagawa 2010) in the assigned category of urbanisation tolerance ($H^2 = 0.693$, CI = 0.497 to 0.861).

Theory suggests a variety of species-level attributes that could explain the varying response of birds to urbanised

environment. Our global analysis yielded no support for some popular hypotheses (Table 1, Tables S5 and S6). Specifically, urban avoiders and exploiters showed no significant differences in terms of the fast–slow continuum, maximum rate of population growth, body size, relative brain size, migratory behaviour, diet type and mating system. However, compared with urban exploiters, urban avoiders tended to exhibit narrower environmental tolerances (measured in terms of habitat breadth and geographic range size), nest in solitary and in exposed sites (e.g. ground) and have a higher brood value (i.e. concentrate the reproductive effort in fewer breeding attempts) (Tables 1, S6).

Having found firm evidence that species vary in tolerance to urbanisation, we finally asked to which extent this accounts for the loss of diversity in urbanised environments. Most species were indeed classified as urban avoiders rather than as urban exploiters (Fig. 3c and d; Table S7). As a consequence, most of the observed loss of species richness in urban habitats was better explained by adaptive differences among species rather than by random processes (Fig. 3a and b).

DISCUSSION

Our global analysis of bird communities confirms that urbanisation causes an important loss of biodiversity, particularly

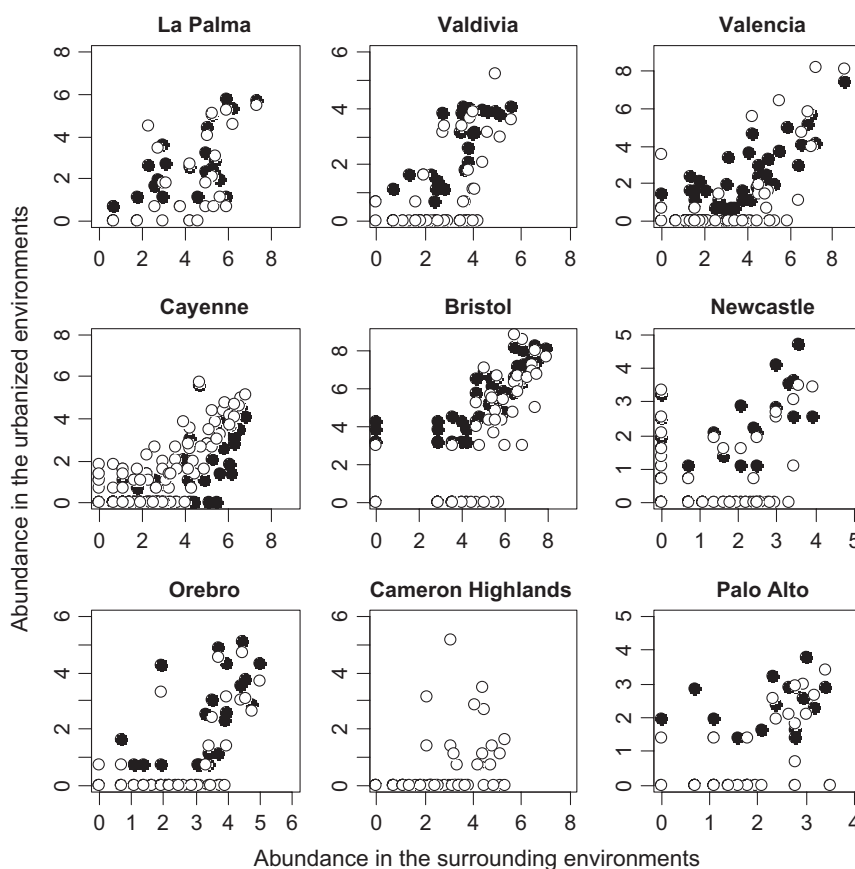


Figure 2 Abundance of avian species in the highly urbanised (white dots) and moderately urbanised (black dots) environments relative to the surrounding more natural environments for a selected number of locations (see Supporting Information Results for the complete comparison). Abundances have been $\log(x+1)$ transformed to reduce the influence of a few large values. Note that for Cameron Highlands only information for the highly urbanised environment was available.

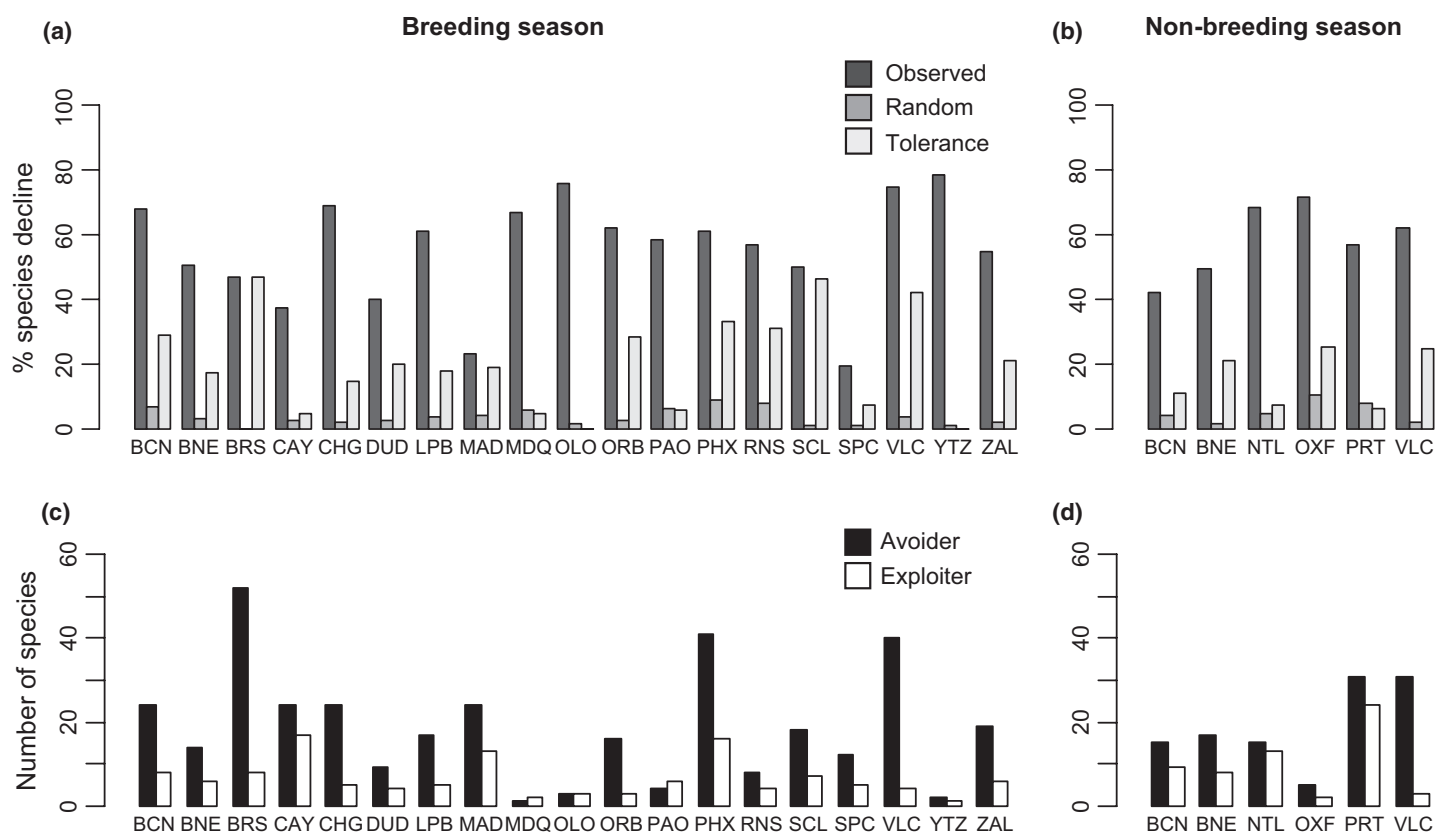


Figure 3 (a and b) Diversity loss observed and expected with urbanisation for the breeding (a) and non-breeding seasons (b) in each location (see abbreviations in Fig. 1). The observed loss is the percentage of species from the surrounding habitats that are absent in the highly urbanised habitats. The expected loss, estimated with simulations (see main text for details), is shown assuming that either random dispersal (Random in the legend) or varying tolerance to urbanisation (Tolerance) is the main driver. (c and d) Number of species classified as urban avoiders or exploiters in each study region according to community simulations (see main text for details) for the breeding (c) and non-breeding seasons (d).

Table 1 Best model accounting for differences between urban avoiders and urban exploiters based on a MCMCglmm including region, species and phylogeny as random factors

	UA vs. UE (<i>n</i> = 358)				RUA vs. UE (<i>n</i> = 268)	
	Post. mean	Lower 95% CI	Upper 95% CI	pMCMC	pMCMC	
Fixed effects						
(Intercept)	-16.44	-33.75	-2.00	0.006		0.024
Brood value	-8.68	-16.75	-1.68	0.002		<0.001
Habitat breadth	6.50	-5.63	17.16	0.214		0.044
Social breeding						
Colonial	0.00	—	—	—		—
Facultative	-1.52	-7.97	4.88	0.654		0.096
Solitary	-5.48	-12.00	-0.26	0.026		<0.001
Nest site						
Ground	0.00	—	—	—		—
Holes	4.54	-1.59	11.20	0.091		0.026
Other	1.69	-3.80	7.17	0.524		0.821
Season						
Non-reproductive	0.00	—	—	—		—
Reproductive	-1.43	-3.51	0.65	0.172		0.034
Random effects						
Location	22.00	2.06	60.75	—		—
Species	32.49	2.73	84.51	—		—
Phylogeny	56.34	0.01	196.10	—		—

Each variable is tested together with the other variables. MCMC *P*-values for fixed effects are also provided for differences between strict urban avoiders and urban exploiters. CI = Confidence interval; UA = urban avoiders; RUA = strict urban avoiders; UE = Urban exploiters.

when urbanisation is intense (see also McKinney 2002, 2006; McDonnell & Hahs 2008; Shochat *et al.* 2010; Aronson *et al.* 2014). Although there is an element of chance in which species are lost, four lines of evidence suggest that most of this loss is associated with a low tolerance of species to urbanisation. First, although urban communities were influenced by the diversity in adjacent landscapes, in agreement with a random dispersal-assembly process, some species were consistently classified across regions as either avoiders or exploiters (see also Møller *et al.* 2012). This supports the fact that species vary in their sensitivity to urbanisation. Second, unlike previous studies (e.g. Evans *et al.* 2011), variation in such sensitivity was non-randomly distributed across the phylogeny. Thus, closely related species tend to respond to urbanisation in a similar way, possibly because they share features that affect their tolerance to urbanisation. The global scale of our analysis has probably contributed to the detection of phylogenetic signal, as there are avian families like *Sturnidae*, *Corvidae* and *Columbidae* that have few urban representatives in local communities but tend to have urban representatives in most regions where they occur. Third, variation among species in urbanisation tolerance was predicted by differences in their life history and the way they use the resources. Finally, in highly urbanised environments, most species were classified as urban avoiders, suggesting that species with attributes to thrive in human-altered habitats are infrequent. Although in ecology there is rarely a one-to-one relationship between patterns and processes (Rosindell *et al.* 2012), all these lines of evidence provide important support for the 'urban tolerance hypothesis' as a major cause of biodiversity loss in cities.

Because urbanisation involves a variety of drastic and rapid environmental alterations, including the replacement and fragmentation of natural vegetation by human structures and increased disturbance from humans and their pets, it is unsurprising that the limits of tolerance of many species are exceeded (Sih *et al.* 2011; Sol *et al.* 2013b). However, what makes a species tolerant to urbanisation is less obvious (Evans *et al.* 2011). According to our analyses, this is primarily related to life history and the way individuals use the resources. Regarding resource use, our results yield some evidence for the hypothesis that urban exploiters have broader environmental tolerance than avoiders, at least in terms of habitat breadth. This is consistent with previous studies (Bonier *et al.* 2007; Evans *et al.* 2011) and agrees with the hypothesis that generalist organisms are more likely to tolerate environmental changes (see references in Bonier *et al.* 2007). However, the relationship was weak and it was non-significant in some models.

Evidence is nonetheless clearer for the importance of life history. Life history integrates key evolved adaptations expected to influence the colonisation-extinction dynamics that ultimately sustain biodiversity (Stearns 2000; Saether *et al.* 2005), and hence has long been considered central to understanding the varying responses of organisms to environmental changes. Surprisingly, however, life-history theory has achieved little success in predicting the response of organisms to novel environments, in part because of the failure to recognise the variety of mechanisms involved (see Sol *et al.* 2012). Our finding that urban exploiters are characterised by a low

brood value (i.e. tend to spread reproductive effort in many breeding attempts) agrees with a number of theories, like bet-hedging, life history-buffering and cognitive-buffer ones, that conceptualise life histories as evolutionary solutions to environmental uncertainties and adaptive mismatches (Sol *et al.* 2012). Urban habitats pose important challenges for reproduction both due to their artificiality and the high frequentation of domestic animals and human beings (see references in Sol *et al.* 2013b), which can explain why we find that in some regions species that nest in solitary or in exposed places tend to avoid cities (Clergeau *et al.* 2006; Evans *et al.* 2011). The inability to spread the risk across several breeding attempts in such novel, uncertain circumstances may thus have important fitness costs, which should increase the probability of local extinction (Williams 1966; Sol *et al.* 2012). On the contrary, distributing reproduction in many events not only reduces the fitness costs of a reproductive failure but can also provide additional benefits. For species that attain a low brood value by extending their reproductive life these benefits include reduced fluctuations of their populations and the possibility to delay reproduction when conditions are unfavourable or when gaining information on the environment is critical (Williams 1966). Species that achieve low brood values by producing several broods per year do not have such benefits, as they typically have shorter reproductive lives, yet they can take advantage of a higher reproductive effort with lower costs of losing a breeding attempt. Thus, although there may be more than one strategy to be a successful urban dweller, our results agree with the recent discovery that rather than investing in high reproductive rates, birds that better cope with novel environmental conditions are characterised by life-history strategies in which they give priority to future rather than current reproduction (Sol *et al.* 2012).

Other attributes classically associated with urbanisation, like the fast-slow continuum, maximum rate of population growth, body size, brain size, migratory behaviour, diet type and mating system, receive little support in our analyses (see also Evans *et al.* 2011). Discrepancies between our analyses and previous studies can reflect differences in approach but also the fact that the importance of most traits is idiosyncratic. Thus, geographic variation in life history and ecological attributes of species in the pool of potential community members often results in responses of communities to local factors (Lessard *et al.* 2012). However, it should be noted that the life history of a species subsumes a number of traits that affect the response to novel stressful conditions, which can obscure the relationship between these traits and urbanisation. Indeed, the brood value concept already integrates the fast-slow continuum (Saether 1988), with high brood values attained by short-lived species that concentrate all reproductions in few attempts and low brood values for long-lived species that distribute the reproductive effort in many attempts. However, the brood value is a broader concept than the fast-slow continuum because as already mentioned a low brood value can be attained not only by a long life but also by reproducing more frequently (Sol *et al.* 2012). Likewise, our results do not contradict that a large brain, and the associated enhanced behavioural flexibility, is important to cope with the challenges of urban environments (Møller 2009; Maklakov

et al. 2011; but see Kark *et al.* 2007). The ability to behaviourally respond to novel ecological challenges can indeed be one of the explanations why species that attain a low brood value through long reproductive lives are so successful in urban environments (Sol *et al.* 2012). However, our results suggest that the issue is more complex than often assumed, and that having a large brain could just be one of a variety of strategies to cope with urbanisation.

Although our analyses suggest that most of the observed loss of species richness in urban habitats is better explained by adaptive differences among species rather than by functional equivalences, there are important discrepancies between the values predicted and those observed. Discrepancies between the rates of species extinction predicted by models and those actually recorded are commonly found in the literature, which has fuelled controversies regarding the real impact of human-induced environmental changes and has led to the proposal of the concept of 'extinction debt' (He & Hubbell 2011). In our case, the discrepancy partially reflects a lack of information. Thus, species that were rarely observed in urban habitats not only included avoiders but also species that were so scarce in the surrounding area that it was not possible to decide whether or not they were tolerant to urbanisation. Moreover, although all cities were founded more than 100 years ago (see Appendix S1), ensuring that most species have had good opportunities for colonisation, the possibility remains that some species classified as urban avoiders are actually recent colonisers that have not yet had enough time to behaviourally and evolutionarily adjust to the new environment. Finally, the discrepancies could also reflect the fact that our approach uses spatial patterns as a proxy for temporal responses and hence cannot capture all the processes involved. A temporal perspective, albeit more difficult given the paucity of data, would be more informative to identify non-linear responses and extinction debts. For the rarest species, more likely to be affected by demographic stochasticity and Allee effects, it may be especially informative to take into account population processes like demographic parameters and species' turnover over time.

The finding that species vary in their tolerance to urbanisation according to their biological traits has significant conservation implications. First, by filtering species according to their biological traits, urban environments reduce β diversity and hence promote biotic homogenisation (Blair 2001; Clergeau *et al.* 2006; Devictor *et al.* 2007; Knapp *et al.* 2008). Second, a disproportionate decline in the abundance of species with particular features could impair the functioning and resilience of ecosystems, altering ecosystem services that are important for human societies (Cardinale *et al.* 2012). Finally, the finding that closely related species exhibit similar sensitivities to human-induced habitat alterations raises the possibility that the urbanisation process drives a disproportionate loss of evolutionary history (Isaac *et al.* 2007). This may happen if vulnerable species are more closely related than expected by chance (Thuiller *et al.* 2011). Future environmental policies and planning should consider the reasons that make highly urbanised environments inhospitable for most species to preserve the elements of natural habitats that, when altered, act as strong environmental filters for most species. As

highlighted by Evans *et al.* (2011), this is difficult in existing urban areas, but may be easier to achieve when designing new urban settlements under future scenarios (e.g. Seto *et al.* 2012) in which the rapid expansion of urban land areas is expected to have a great impact on global biodiversity.

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DS, CGL and DM conceived the study, all authors contributed to data gathering, DS, CGL, DM and OL performed the comparative analyses, DS and JM ran the simulations, DS and CGL wrote the manuscript and all authors contributed to the different drafts of the manuscript.

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